MACHINE LEARNING PROJECT

Customer Purchase Prediction

Using Logistic Regression and Random Forest

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PREDICT • COMPARE • LEARN

Introduction

Project background and need

What is it?

An ML application that estimates whether a customer may purchase a product.

Input data

Customer age and estimated salary are the input features.

Why needed?

Supports marketing planning and customer analytics.

Domain: Machine Learning | Classification | Customer Analytics

The system compares two classification algorithms on the same dataset.

Problem Statement & Motivation

Why customer purchase prediction matters

Problem

- Purchase decisions are uncertain.
- Manual targeting may waste time and budget.
- Businesses need data-supported insights.

Motivation

- Explore a practical ML use case.
- Compare a linear classifier with a tree ensemble.
- Demonstrate predictions through a browser interface.

Potential users: retailers, e-commerce teams, marketers and customer analytics teams.

Project Objectives

What the project aims to achieve

01 **Prepare data**

Load, validate and explore customer records.

02 **Train models**

Build two classification models.

03 **Evaluate**

Measure performance on test data.

04 **Predict**

Estimate purchase likelihood from new inputs.

05 **Compare**

Present results in an interactive interface.

Existing System vs Proposed System

From manual decisions to model-assisted predictions

Existing approach

- Manual customer segmentation
- Decisions based on experience or simple rules
- Difficult to compare methods consistently
- Limited automated prediction

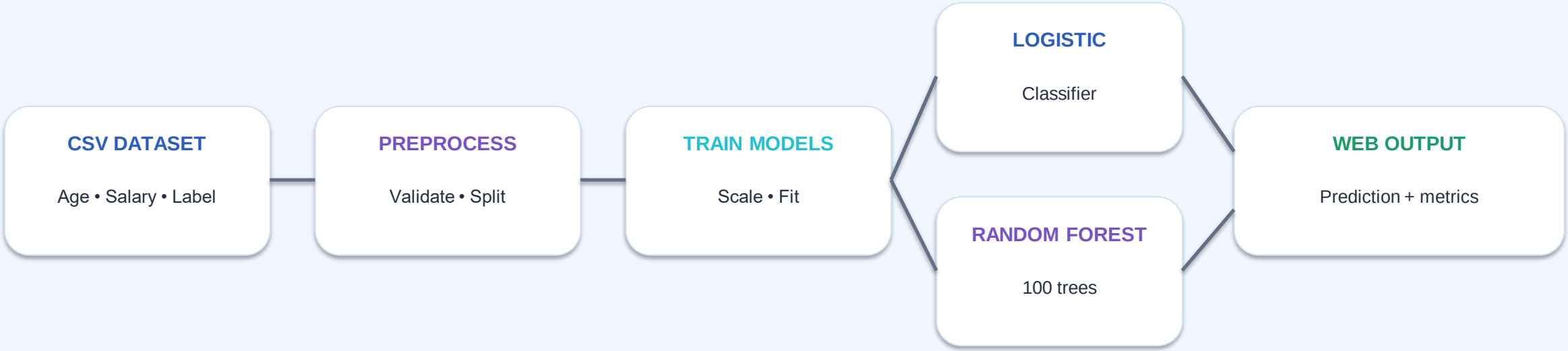
Proposed approach

- Train two ML classifiers on labelled data
- Use age and salary as inputs
- Evaluate with held-out test data
- Show prediction and probability in a web interface

The system supports decisions; it does not guarantee customer behavior.

System Architecture

Main components and data flow



The user enters Age and Salary; trained models return a class prediction and estimated probability.

Methodology / Workflow

End-to-end project pipeline

1. Load CSV
2. Validate data
3. Explore & clean
4. Split 80/20
5. Train models
6. Evaluate
7. Predict new input
8. Display in browser

Test data is kept separate from training data to estimate performance on unseen records.

Dataset & Input Features

Data used to train the classifiers

Feature	Description	Example
Age	Customer age	35
EstimatedSalary	Estimated salary	₹60,000
Purchased	Target label	0 = No, 1 = Yes
Source	CSV file	customer_purchase_dataset.csv

Inputs (X)

Age, EstimatedSalary

Target (y)

Purchased: 0 or 1

Check missing values, duplicates, column names and class balance before training.

Algorithms Used

Two supervised classification models

Logistic Regression

- Estimates class probability.
- Uses scaled input features.
- Provides probability for Purchased = 1.
- Useful as a baseline classifier.

Random Forest

- Combines multiple decision trees.
- Can learn non-linear relationships.
- Uses 100 trees in this project.
- Provides class predictions and probabilities.

Both models use the same train/test split for a fair comparison.

Technology Stack

Software and tools

Python

Main programming language

Pandas / NumPy

Data handling and numerical operations

Scikit-learn

Preprocessing, training and evaluation

Matplotlib

Charts and confusion matrices

Gradio

Interactive browser interface

Google Colab

Cloud notebook and execution

A database is not required for this prototype; records are stored in a CSV file.

Implementation & Web Demo

How users interact with the application

1. Load dataset

Upload a CSV file and preview customer records.

2. Enter details

Example input:

Age = 35

Estimated Salary = ₹60,000

3. View prediction

Display predictions, probabilities and test accuracy for both models.

Add screenshots from your actual running Colab/Gradio interface before submission.

Results & Evaluation

Metrics are calculated from held-out test records

Accuracy

Fraction of correct predictions.

Precision

Of predicted purchases, fraction actually positive.

Recall & F1

Positive cases found; F1 balances precision and recall.

Model comparison — fill with actual Colab results

Logistic Regression		**Actual accuracy: _____ %**
Random Forest		**Actual accuracy: _____ %**

Use actual calculated metrics; do not report invented results.

Advantages, Applications & Limitations

Scope of the prototype

Advantages

- Fast predictions
- Compare two models
- Simple browser interface
- Reproducible workflow

Applications

- Retail and e-commerce
- Marketing planning
- Customer analytics
- ML classification demos

Limitations

- Only two input features
- Synthetic data may not reflect reality
- Depends on data quality
- Predictions are not guarantees

Future Scope

Possible extensions

1. **More features** — Purchase history, product category, visits and promotions.
2. **Real dataset** — Use representative customer data with appropriate permissions.
3. **Model improvements** — Cross-validation, tuning and threshold analysis.
4. **Deployment** — Host on a persistent web platform.
5. **Monitoring** — Track drift, performance and fairness.

CONCLUSION

From customer data to purchase insights

AGE

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SALARY

This project demonstrates dataset preparation, model training, evaluation and interactive prediction in a browser-based interface.

THANK YOU